

Q1 - 25 July - Shift 1

A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that $QR = 15$ m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is :

- (A) $5(2\sqrt{3} + 3)$ m (B) $5(\sqrt{3} + 3)$ m
(C) $10(\sqrt{3} + 1)$ m (D) $10(2\sqrt{3} + 1)$ m

Space for your notes:

Q2 - 27 July - Shift 1

Let a vertical tower AB of height $2h$ stands on a horizontal ground. Let from a point P on the ground a man can see upto height h of the tower with an angle of elevation 2α . When from P, he moves a distance d in the direction of $\overrightarrow{AP}$, he can see the top B of the tower with an angle of elevation α . If $d = \sqrt{7}h$, then $\tan \alpha$ is equal to

- (A) $\sqrt{5} - 2$ (B) $\sqrt{3} - 1$
(C) $\sqrt{7} - 2$ (D) $\sqrt{7} - \sqrt{3}$

Space for your notes:

Q3 - 27 July - Shift 2

Questions

MathonGo

The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60° . If $\angle BAQ = 30^\circ$, $AB = d$ and the area of the trapezium PQRB is α , then the ordered pair (d, α) is :

- (A) $(10(\sqrt{3}-1), 25)$ (B) $(10(\sqrt{3}-1), \frac{25}{2})$
(C) $(10(\sqrt{3}+1), 25)$ (D) $(10(\sqrt{3}+1), \frac{25}{2})$

Space for your notes:

Q4 - 28 July - Shift 2

A horizontal park is in the shape of a triangle OAB with $AB = 16$. A vertical lamp post OP is erected at the point O such that $\angle PAO = \angle PBO = 15^\circ$ and $\angle PCO = 45^\circ$, where C is the midpoint of AB. Then $(OP)^2$ is equal to

- (A) $\frac{32}{\sqrt{3}}(\sqrt{3}-1)$ (B) $\frac{32}{\sqrt{3}}(2-\sqrt{3})$
(C) $\frac{16}{\sqrt{3}}(\sqrt{3}-1)$ (D) $\frac{16}{\sqrt{3}}(2-\sqrt{3})$

Space for your notes:

Q5 - 29 July - Shift 1

#MathBoleTohMathonGo

Questions

MathonGo

The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot \alpha$ is equal to :

(A) $\frac{6}{5}$

(B) $\frac{9}{5}$

(C) $\frac{4}{3}$

(D) $\frac{7}{3}$

Space for your notes:

#MathBoleTohMathonGo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Answer Key

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q1 (A)

Q2 (C)

Q3 (A)

Q4 (B)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

Q5 (A)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

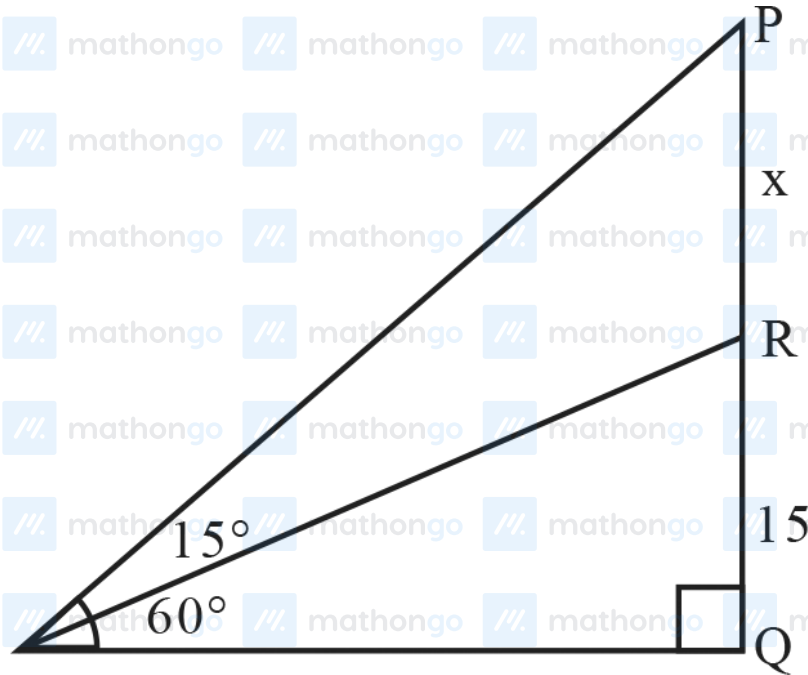
// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

#MathBoleTohMathonGo

Q1 (A)



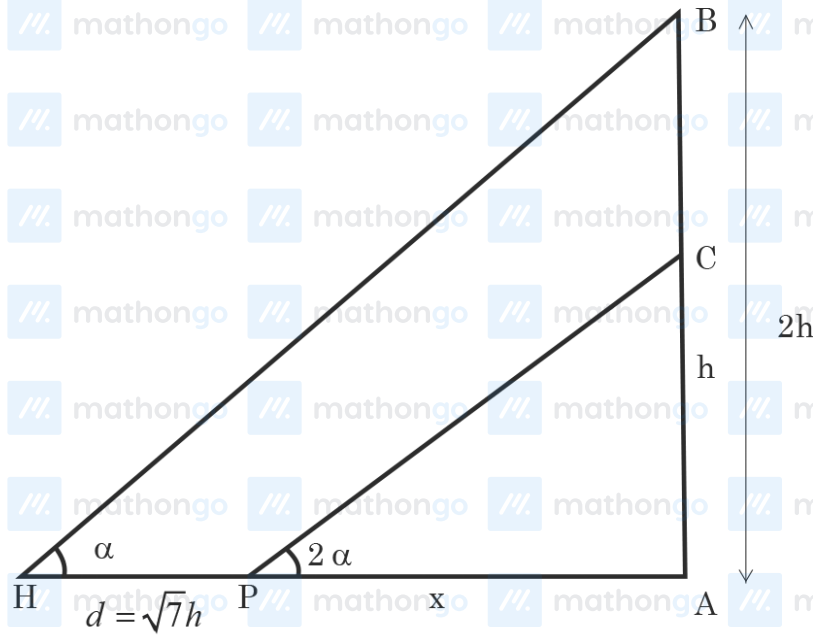
$$\frac{15}{AQ} = \tan 60^\circ \quad \dots(1)$$

$$\frac{15 + x}{AQ} = \tan 75^\circ \quad \dots(2)$$

$$\frac{(1)}{(2)} \Rightarrow x = 10\sqrt{3}$$

$$\text{So, } PQ = 5(2\sqrt{3} + 3) \text{ m}$$

Q2 (C)



$$\tan 2\alpha = \frac{h}{x}$$

$$\text{and } \tan \alpha = \frac{2h}{x + \sqrt{7}h}$$

$$\tan \alpha = \frac{2h}{h \cot 2\alpha + \sqrt{7}h}$$

$$\tan \alpha = \frac{2}{\frac{(1 - \tan^2 \alpha)}{2 \tan \alpha} + \sqrt{7}}$$

Put $\tan \alpha = t$ & simplify

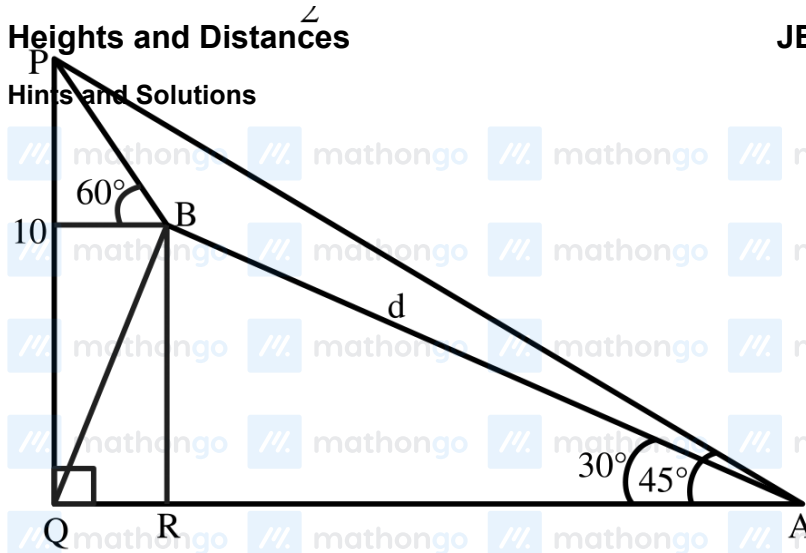
$$\Rightarrow \tan \alpha = \sqrt{7} - 2$$

Q3 (A)

$$QA = 10 \quad RA = d \cos 30^\circ = \frac{\sqrt{3}d}{2}$$

$$QR = 10 - \frac{\sqrt{3}d}{2}$$

#MathBoleTohMathonGo



$$\tan 60^\circ = \frac{PQ - BR}{QR} = \frac{10 - \frac{d}{2}}{10 - \frac{\sqrt{3}d}{2}}$$

$$\Rightarrow \sqrt{3} = \frac{20 - d}{20 - \sqrt{3}d}$$

$$\Rightarrow 20\sqrt{3} - 3d = 20 - d$$

$$\Rightarrow 2d = 20(\sqrt{3} - 1)$$

$$\Rightarrow d = 10(\sqrt{3} - 1)$$

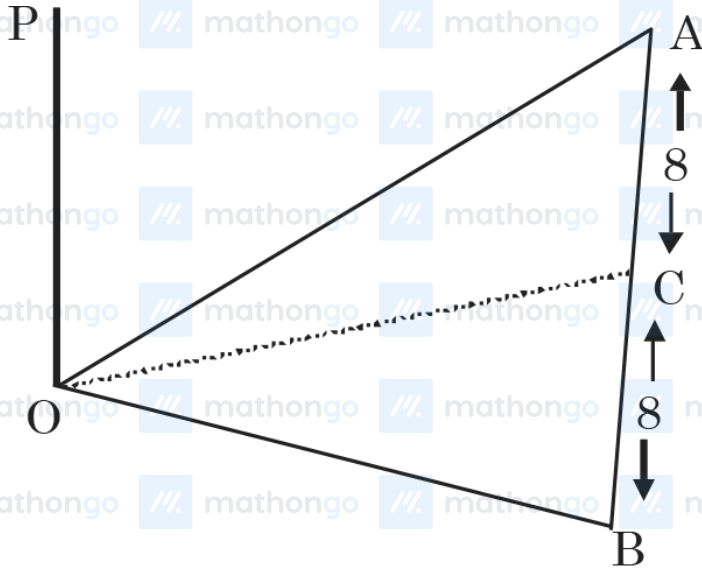
$$\text{ar(PQRB)} = \alpha = \frac{1}{2}(PQ + BR) \cdot QR$$

$$= \frac{1}{2} \left(10 + \frac{d}{2} \right) \cdot \left(10 - \frac{\sqrt{3}d}{2} \right)$$

$$= \frac{1}{2} (10 + 5\sqrt{3} - 5) (10 - 15 + 5\sqrt{3})$$

$$= \frac{1}{2} (5\sqrt{3} + 5) (5\sqrt{3} - 5) = \frac{1}{2} (75 - 25) = 25$$

Q4 (B)



Sol.

$$\frac{OP}{OA} = \tan 15^\circ$$

$$\Rightarrow OA = OP \cot 15^\circ$$

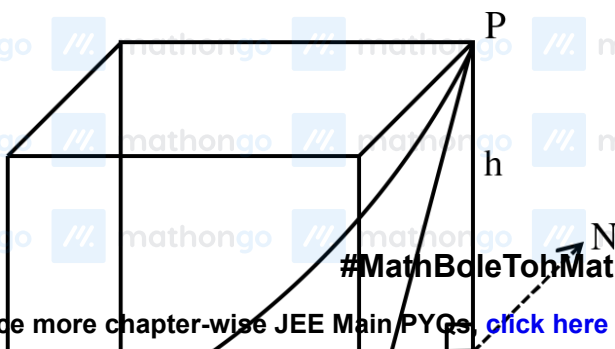
$$\frac{OP}{OC} = \tan 45^\circ \Rightarrow OP = OC$$

$$\text{Now, } OP = \sqrt{OA^2 - 8^2}$$

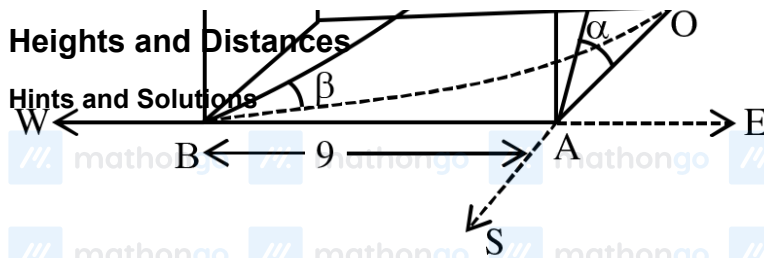
$$\Rightarrow OP^2 = (OP)^2 \cot^2 15^\circ - 64$$

$$\Rightarrow OP^2 = \frac{32}{\sqrt{3}}(2 - \sqrt{3})$$

Q5 (A)

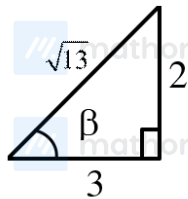


#MathBoleToMathonGo

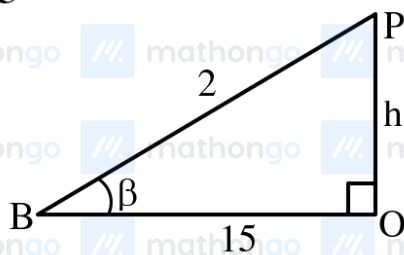


given $OB = 15$

$$\cos \beta = \frac{3}{\sqrt{13}}$$



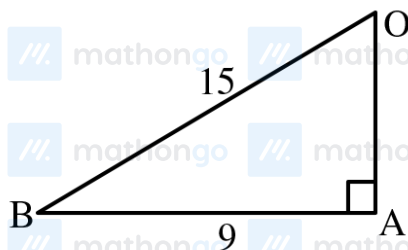
$$\tan \beta = \frac{2}{3}$$



$$\tan \beta = \frac{h}{15}$$

$$\frac{2}{3} = \frac{h}{15}$$

$$\boxed{10 = h}$$

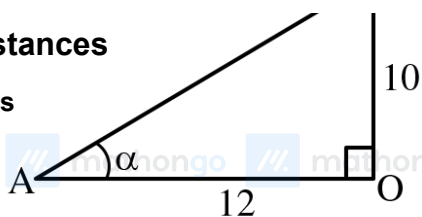


$$OA^2 + AB^2 = 225$$

$$OA^2 + 81 = 225$$

$$\boxed{OA = 12}$$

#MathBoleTohMathonGo



$$\tan \alpha = \frac{10}{12}$$

$$\cot \alpha = \frac{12}{10} = \frac{6}{5}$$